

Hub SSoT — How the Governed-RAG Problem Is Solved End to End

A single, governed, agentic Source of Truth for Telefónica's Communication & Brand teams. This document explains — grounded in the actual implementation — how Hub SSoT answers the central question of the PC2 / RFP synthesis: **how do you let an AI answer over a sensitive corpus without ever letting it leak, guess, or invent?**

The short answer, and the design principle behind every section that follows:

Governance decisions are made deterministically, before the model. The LLM only ever composes prose from evidence the user was already permitted to read. It never filters, never decides access, and never sees blocked content.

1. The problem being solved

Telefónica's Communication, Brand and Cabinet teams sit on a corpus that mixes public press material with private briefings, confidential strategy and off-the-record board material. The RFP asks for a system where:

- People ask questions in natural language and get answers **always backed by cited evidence**.
- Access rights are the **intersection of two axes** — user (area/profile) × document (confidentiality label) — **applied at index/retrieval time (early binding), never by the agent**. If a user lacks permission, a chunk must never enter the model context.
- Taxonomy and tagging are **configuration, not code**: embeddings are computed once and stay agnostic to the taxonomy, so a strategy change re-tags existing data without re-ingestion, re-embedding, redeployment — or IT.
- When a confident, permitted answer is not possible, the system gives an **honest refusal** — "no evidence", "permission blocked", or "this source is historic" — rather than a plausible hallucination.

A conventional RAG pipeline fails all three: it filters (if at all) after retrieval, trusts the model to be discreet, re-embeds when metadata changes, and answers even when it should not. Hub SSoT is built so those failure modes are structurally impossible, not just discouraged by a prompt.

2. Architecture at a glance

The demo is a pnpm monorepo with a contract-first API:

- **Governed corpus** — a synthetic but fully governed Telefónica corpus: internal (A), external (B) and SSOT-generated (E) documents, each carrying country, brand, legal entity, period, type, confidentiality (**public** / **private** / **confidential** / **off_the_record**), owner, validity (**approved** / **historic** / **review** / **superseded**) and area scope (Comunicación, Marca, Gabinete). External sources carry a pre-ingest filter record (keywords, competitors, executives, topics, sentiment) — filtered before ingestion, never a raw dump.
- **Vector index** — hybrid retrieval (dense semantic vectors + sparse BM25 from one shared multilingual tokenizer) with all governance labels stored as point payload. Runtime-ingested documents persist their full governed payload, so a restart rebuilds the corpus from the index.
- **One access resolver** — a single **area × clearance** intersection rule (**resolveDocAccess**) that every retrieval path, side channel, downstream flow and export gate must use. Clearance is checked first, then area scope.
- **Agents** — Ask (governed Q&A), Generate (governed content creation), Brand Guardian (two-pass review), KPI chat (governed figures), a schedule runner (recurring drafts with human approval) and an export service (final server-side gates). Composition uses Claude behind Lyzr-named adapter interfaces (**kb**, **kg**, **numeric**, **text**), so a real Lyzr backend can be swapped in without touching the agents or routes.
- **Audit** — every retrieval, on every surface, is logged with the persona, the filters applied, every hit and its access decision, and the final outcome.

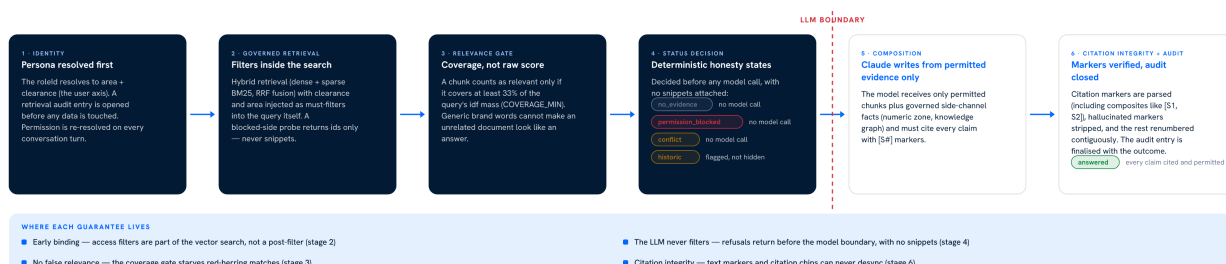
Persona/clearance is client-asserted in this demo (a persona switcher, not real authentication); everything downstream of that assertion is enforced exactly as the target architecture requires.

3. The complete Ask RAG pipeline

HUB SSOT · ASK WORKFLOW

The complete Ask RAG pipeline — question to cited answer

Every governance decision is made deterministically, in order, BEFORE the model boundary. The LLM only ever composes prose from chunks the persona was already permitted to read — it never filters, never decides access, and never sees blocked content.



Each stage is an actual code path: governance resolver, governed retrieval front door, coverage gate, status decision and citation pipeline in the ask agent.

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The complete Ask RAG pipeline

A question travels through six ordered stages. The first four are deterministic code; the model appears only at stage five, and only when a permitted answer is possible.

Stage 1 — Identity first. The persona's **roleid** resolves to an area and a clearance level — the user axis. A retrieval audit entry is opened before any data is touched. Permissions are re-resolved on every conversation turn, and conversation memory is persona-scoped: history is filtered by the current role so a lower-clearance persona can never inherit a higher-clearance turn.

Stage 2 — Governed retrieval. The query runs as a hybrid search (dense + sparse, reciprocal-rank fusion) with the persona's clearance and area injected as **must-filters inside the query itself**. This is early binding as PC2 defines it: the governance filter **is** the search, not a post-filter applied by an agent. A separate blocked-side probe returns **ids only** — used solely to distinguish "nothing exists" from "something exists that you cannot see", never to fetch snippets — and a live-label re-check on those ids fails closed.

Stage 3 — Relevance by coverage, not raw score. A chunk only counts as relevant if it covers at least 33% of the query's idf mass (**COVERAGE_MIN**). This is a deliberate departure from absolute BM25 thresholds: generic brand words ("Telefónica") or stray verbs ("strategy") carry little idf mass, so they cannot make an unrelated public document look like an answer. The gate is what makes **no_evidence** trustworthy.

Stage 4 — Deterministic status decision. Before any model call, the pipeline decides the honesty state (section 6). **no_evidence** and **permission_blocked** return immediately, carrying **no snippets — only a classification label**. Conflicts between the top two permitted sources are surfaced, not reconciled. Superseded evidence flags the answer as historic.

Stage 5 — Composition. Only now does Claude see anything: the permitted chunks, plus governed side-channel facts (the numeric zone and the knowledge graph, both access-checked through the same resolver). The system rules require every claim to carry an **[S#]** citation marker and forbid outside knowledge.

Stage 6 — Citation integrity and audit. The response is parsed for citation markers — including composite markers like **[S1, S2]** — hallucinated or stray markers are stripped, and the survivors are renumbered contiguously so inline markers and citation chips can never desync. The audit entry is finalised with the outcome.

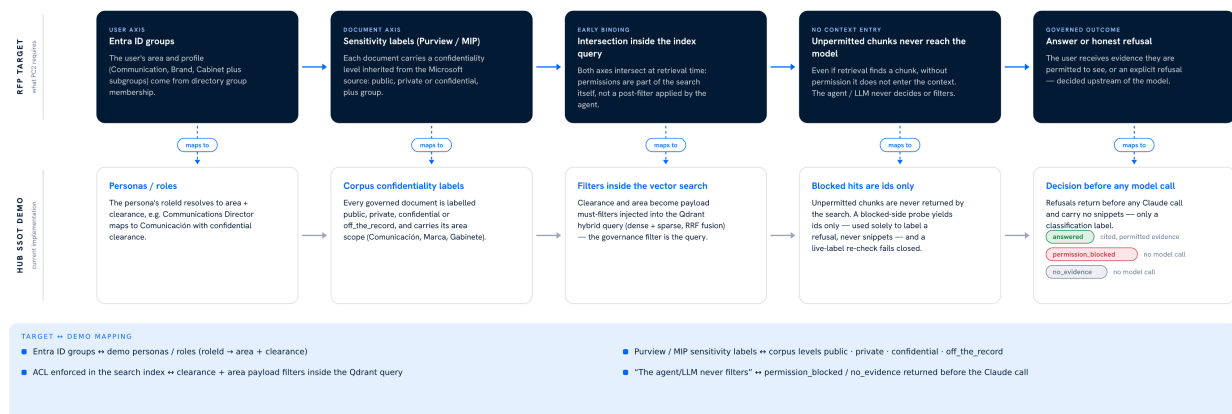
For follow-up turns that initially find nothing, a small router model classifies the intent; genuine topic questions get one second chance through the **same** governed retrieval with a context-resolved standalone query. The router changes the query, never the rules — and non-topical wording (chat phrasing, refine instructions) is never folded into the coverage-gated query, so it cannot dilute the relevance ratio.

4. Early-binding access control

PC2 - ACCESS RIGHTS

Access rights — early-binding access control

Access = the intersection of two axes (user axis × document axis), applied at index / retrieval time — never filtered by the agent or the LLM. If a user lacks permission, a chunk never enters the model context.



Every demo component corresponds to an actual code path in the Hub SSoT implementation (governance resolver, governed retrieval front door, ask agent).

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Access rights — early-binding access control

This is the existing PC2 access-rights diagram; the solution described here is the demo lane of that diagram, unchanged.

PC2's rule: *access = intersection of two axes, applied at the index/retrieval (early-binding), never by the agent.* The mapping in the implementation:

PC2 target	Hub SSoT implementation
User axis from Entra ID groups	Persona roleid → area + clearance via the governance resolver
Document axis from Purview/MIP sensitivity labels	Corpus confidentiality labels public / private / confidential / off_the_record + area scope
Intersection inside the index query	Clearance + area payload must-filters inside the hybrid vector search
Unpermitted chunks never reach the model	Blocked-side probe yields ids only; refusals return before the Claude call with no snippets

Three properties make this early binding real rather than nominal:

- 1. The filter is part of the search.** There is no code path where an unpermitted chunk is retrieved and then discarded by agent logic — the index never returns it.
- 2. One resolver, no side doors.** Numeric facts, knowledge-graph traversal, KPI figures and export citation checks all resolve access through the same **area × clearance** rule. (An earlier iteration had a rank-only check on one side channel; it was found and closed — the lesson is codified: every side channel gates through the shared resolver.)
- 3. Fail closed.** A numeric fact whose source document is missing is treated as inaccessible. A blocked-probe id whose live label cannot be re-verified is treated as blocked.

5. Taxonomy and tagging as configuration

Taxonomy and tagging — configuration, not code

Two taxonomy layers, versioned as configuration and applied after embedding. Embeddings are taxonomy-agnostic, so re-tagging never requires re-ingestion, re-embedding, redeployment — or IT.



Every demo component corresponds to an actual code path in the Hub SSoT implementation (governance engine, re-tag agent, vector store payload updates).

Telefonica · Hub SSoT

Taxonomy and tagging — configuration, not code

This is the existing PC2 taxonomy diagram; again, the demo lane is exactly what is implemented.

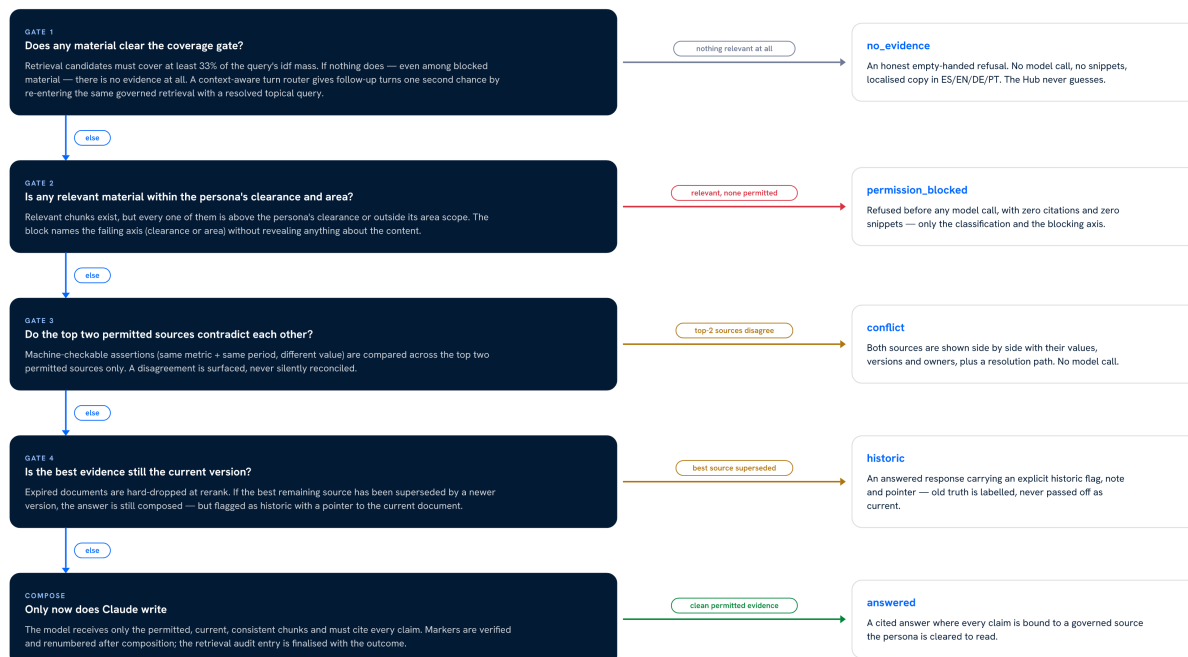
PC2's anchor: **"changing the strategy = a governance operation, not a technical project."** The taxonomy has two layers — deterministic (what the document **is**: market, brand, owner, confidentiality, validity) and derived (what it **says**: plan axes, topics — the layer that changes with strategy). The derived layer is versioned configuration, never code:

- The seed corpus ships classified against taxonomy **v4**; every applied edit creates v5, v6, ... with actor, note and audit trail, persisted and re-applied at boot.
- The governed re-tag flow is the same four steps PC2 specifies: **axis edit** in the governance panel → **candidate mapping table** of affected documents → **assisted zero-shot re-classification** (Claude proposes) → **human validation** (a person accepts or rejects each proposal). Only human-accepted decisions become overrides in a committed version.
- Applying a version mutates **metadata only** — axis labels, **axisids**, **topics** — as payload updates on existing index points. Chunks, embeddings and the retrieval index are untouched: no re-ingestion, no re-embedding, no redeploy, no IT.
- The index commits first; only then is the version committed locally — so the vector store and the taxonomy version can never disagree in the dangerous direction.

6. The honesty states

How the answer status is decided — before composition

The five states are decided deterministically, in this order, before the model writes a word. Blocked and empty outcomes carry no snippets — only a classification label — so a refusal can never leak the content it refuses to show.



Statuses: answered · no_evidence · permission_blocked · conflict · historic — decided deterministically in the ask agent, upstream of composition.

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How the answer status is decided

Where most RAG systems have one output (an answer), Hub SSoT has five, decided in a strict order **before composition**:

Status	Meaning	Model called?	Snippets attached?
no_evidence	Nothing in the governed corpus covers the question	No	None
permission_blocked	Relevant material exists; none of it is within the persona's clearance/area	No	None — only the blocking axis
conflict	The top two permitted sources disagree on the same metric and period	No	Both sources, side by side
historic	Answered, but the best source has been superseded	Yes	Flagged, with a pointer to the current document
answered	Cited answer from permitted, current, consistent evidence	Yes	Every claim cited

Two details matter for trust:

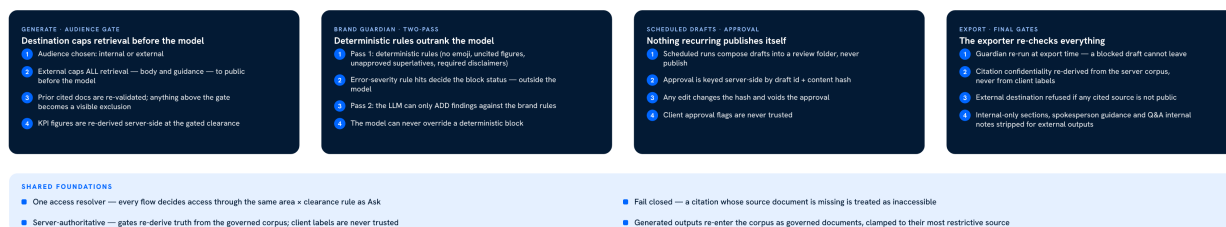
- **Refusals cannot leak.** A `permission_blocked` response carries zero snippets and zero citations — it names the failing axis (clearance or area), nothing else. The refusal also carries a machine-readable code, so the frontend keys its remediation UX ("switch persona", "request access") on the code rather than parsing prose.
- **Conflicts are surfaced, never merged.** Conflict detection compares machine-checkable assertions (same metric + same period, different value) across the top two permitted sources only — retrieval noise further down the ranking cannot fabricate a conflict, and corroboration ("n sources agree") counts distinct cited documents, not retrieved chunks.

7. Downstream flows inherit the same guarantees

HUB SSoT · DOWNSTREAM GOVERNANCE

The same guarantees downstream — generation, export, review

Everything built on top of retrieval re-applies the same rules server-side: permissions resolve before retrieval, destination confidentiality caps what the model can see, deterministic rules outrank the model, and approvals are server authoritative.



The same guarantees downstream

Answering questions is only half the product; the other half is generating governed content. Every downstream flow re-applies the same principles server-side.

Generate — the audience gate. When a draft's audience is **external**, the effective clearance for **all** retrieval — body sources **and** internal guidance material — is capped to **public before** the model runs. Retrieval still runs at the persona's real clearance in parallel, but only to explain what was excluded: anything above the gate appears as a visible exclusion, never as content. KPI figures referenced in a brief are re-derived server-side at the gated clearance, not copied from the client.

Brand Guardian — two passes, deterministic rules win. Pass one is deterministic: hard rules (no emoji, uncited figures, unapproved superlatives, required disclaimers) scan the text, and any error-severity hit decides the **block** status — outside the model. Pass two asks Claude to review against the brand rulebook, but the model can only **add** findings; it can never override a deterministic block. Quotes from the model are mapped back to exact text spans deterministically.

Scheduled drafts — server-authoritative approval. Recurring runs compose drafts into a review folder; nothing recurring publishes itself. Approval is registered server-side, keyed by **draft id + content hash** — any edit to an approved draft changes the hash and silently voids the approval. Client-side approval flags are never trusted.

Export — the final gate chain. The exporter re-checks everything at the moment of export: the Guardian is re-run (a blocked draft cannot leave), scheduled approval is re-verified against the hash registry, press releases require an editorial-review record, and the destination gate **re-derives the confidentiality of every citation from the server corpus** — never from labels the client sent. An external export with any non-public cited source is refused with [confidentiality_blocked](#). Internal-only sections, spokesperson guidance and Q&A internal notes are stripped from external outputs on the server.

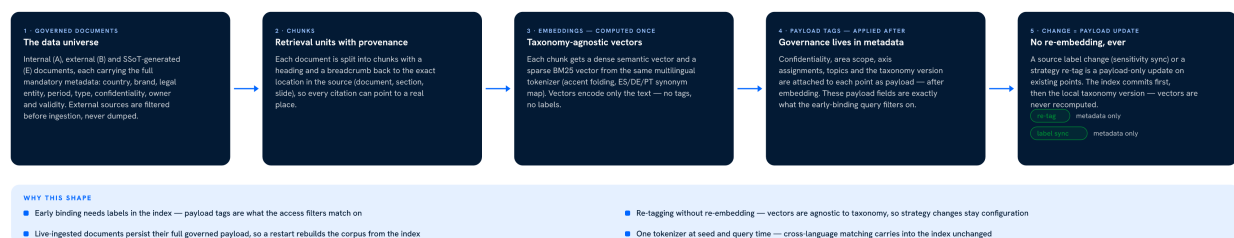
Finally, generated outputs re-enter the corpus as governed **E-category documents** with lineage back to their sources, clamped to the most restrictive confidentiality among them — so the SSoT's own outputs are governed by the same rules that produced them.

8. Corpus and indexing — embeddings once, tags after

HUB SSoT · CORPUS & INDEX

Documents to index — embeddings once, tags after

The index is built so that governance and taxonomy live entirely in metadata. Vectors are computed once and never again: access-label changes and strategy re-tags are payload updates, not re-ingestion, re-embedding or redeployment.



Each stage is an actual code path: the corpus model, chunk registry, seed pipeline, sparse/dense encoders and the payload update operations in the vector-store adapter.

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Documents to index — embeddings once, tags after

The index is shaped by one requirement: governance and taxonomy must live entirely in metadata.

- 1. Governed documents** carry the full mandatory metadata (section 2). External (B) documents additionally record what the pre-ingest filter matched.
- 2. Chunks** carry a heading and a breadcrumb back to the exact location in the source (document › section › slide), so every citation points to a real place a human can verify.
- 3. Embeddings are computed once.** Each chunk gets a dense semantic vector and a sparse BM25 vector from the same multilingual tokenizer (accent folding first, then a curated EN/ES/DE/PT synonym map). The vectors encode only the text — no tags, no labels.
- 4. Payload tags are applied after embedding:** confidentiality, area scope, [axisIds](#), topics, validity and the taxonomy version. These payload fields are exactly what the early-binding query filters match on.
- 5. Every subsequent change is a payload update.** A sensitivity-label change or a strategy re-tag is [set_payload](#) on existing points; vectors are never recomputed. Live-ingested documents persist their full governed payload in the index, so a restart hydrates the corpus from it.

This is the structural reason the taxonomy promise holds: because the vectors never encoded the taxonomy in the first place, there is nothing to re-embed when the taxonomy changes.

9. Auditability

Every retrieval, on every surface (Ask, Generate, KPI chat), writes an audit entry containing the surface, persona ([roleid](#), label, clearance, area), the query, the exact filter expression applied, every hit with its score and access decision, and the final status. Entries are persisted to disk with a debounced writer and are queryable by document or persona — so a governance owner can answer both "who retrieved this document?" and "what did this persona see?" after the fact. The entry is opened **before** retrieval and finalised with the outcome, so even refused turns leave a trace.

10. What is real and what is faithfully simulated

The demo is honest about its own boundaries:

Real, working code	Faithfully simulated
Early-binding filters inside the hybrid vector search	Entra ID group membership (demo personas assert the role)
The full honesty-state decision pipeline, pre-model	Purview/MIP label inheritance (labels are seeded on the corpus)
Versioned taxonomy with human-validated re-tagging, payload-only	Source connectors (SharePoint sync, API feeds are labelled provenance)
Claude composition with citation verification and renumbering	The corpus content itself (synthetic, clearly marked fictional)
Guardian two-pass, schedule approval registry, export gate chain	Lyzr platform (native engine behind Lyzr-named adapter interfaces)
Retrieval audit log with per-hit access decisions	Real authentication (persona switching is client-asserted)

The swap path is explicit: identity plugs in at the single access resolver, real sensitivity labels plug in at the document metadata, and a real Lyzr backend plugs in behind the existing [kb](#) / [kg](#) / [numeric](#) adapter interfaces — none of which requires touching the agents, the gates or the routes.

Appendix

- **RAG coverage matrix** — the detailed requirement-by-requirement mapping of PC2 / RFP synthesis items to implementation status, kept alongside this document.
- **Reference diagrams** — the two PC2 requirement diagrams embedded in sections 4 and 5 live in [exports/diagrams/](#) ([access-rights-early-binding](#), [taxonomy-as-configuration](#)), together with the four workflow diagrams introduced by this document ([ask-rag-pipeline](#), [honesty-states-decision](#), [downstream-governed-flows](#), [corpus-indexing-pipeline](#)), each available as SVG and PNG.